

Probing the trivial vacuum of the BMN model

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Abstract

The matrix model on a plane-wave background is discussed in the classical regime. We propose a geometry for the trivial vacuum motivated by a specific limit that arises from the M-theory description where the data of M-theory can be mapped to an electrostatic problem in three dimensions. Furthermore, we investigate the trivial vacuum of the BMN model and give a gravity description in terms of D0-branes forming and probing this geometry. Using a single D0-brane as a probe, we find a repulsive behavior as approaching the core of this geometry. We comment on implications of this believing also that it lies at the reach of Monte-Carlo simulations.

1 Introduction

Matrix models have played a significant role in the development of quantum gravity theories. In a sense it seems that they can lead to emergent (quantum) gravity theories [1], [2], [3], [4]. In this work we focus on matrix models motivated by M-theory [5], [6]. These models are believed to give a non-perturbative *definition* of string and M-theory. In particular, it seems that they capture more characteristics of quantum gravity than string theory, and for this reason they are believed to be more fundamental than string theory itself [7].

Even though there is a plethora of these, here we will focus in the BMN model [6]. The main idea of these models is the following: one starts from a two-sided approach, one of which is M-theory and the other one is Yang-Mills theory. M-theory naturally lives in a eleven-dimensional spacetime, while on the other hand Yang-Mills theories can be consistently written in any dimension. Accordingly one can write a ten-dimensional analogue of Yang-Mills. Starting from the gauge theory side, the recipe indicates to compactify nine out of ten dimensions of the Yang-Mills theory, such that in the end one is left with just one dimension which is nothing more than $(0 + 1)$ -quantum mechanics. Since it is somewhat awkward to have one spatial dimension and study dynamics of the system, usually this dimension is connected to time. The variables in this model are Hermitian matrices X^i where $i = 1, \dots, 9$. In a sense, the gauge fields living in the ten-dimensional spacetime upon compactification they become scalar fields in the low dimensional spacetime. Then for each compactified dimension there is an index i for the scalar field. This is the Yang-Mills side.

From the M-theory perspective we have the following procedure: we compactify just one dimension out of eleven which we might take to be a spatial circle of radius R_s . It is understood from the very early work of Kaluza [8] and Klein [9], that compactification on a spatial dimension on a circle gives us for free some modes, the so-called Kaluza-Klein modes. These, appear in the lower dimensional spacetime and physically correspond to (longitudinal) momenta running in the compactification circle given by the following relation

$$p = \frac{N}{R_s},$$

where N is the number of modes. In a rough sense one can think of this, as a mechanism that produces mass in a lower dimensional spacetime by compactifying massless modes in spatial dimensions of a higher dimensional spacetime. In our case, one unit of mass is $\frac{1}{R_s}$.

In addition, one has to perform a boost with boosting parameter

$$\gamma = \sqrt{\frac{R^2}{2R_s} + 1}, \tag{1.1}$$

and take the limit $R_s \rightarrow 0$. Physically, this means that we boost along the compactification circle whose radius goes to zero. The result of this is that γ becomes infinite corresponding to a lightlike circle of radius R [10]. This procedure indicates a duality between boosted spatial compactifications with $R_s \rightarrow 0$ and lightlike compactifications. However, in this way we generate closed lightlike curves. At the end of the day to talk about physical implications, one has to take the $R \rightarrow \infty$ limit. In other words, the physically relevant limit is

$$p = \frac{N}{R} = \text{fixed} \quad , \quad N \rightarrow \infty \quad , \quad R \rightarrow \infty. \tag{1.2}$$

Furthermore, we remind that the compactified M-theory along the circle with R_s radius, where $R_s = \sqrt{\alpha'} g_s$ becomes ten dimensional in the $R_s \rightarrow 0$ limit. In fact this is believed to be the type IIA string theory and actually it is used as the definition of it. Moreover, in this limit the low energy sector which consists of D0-branes decouples. One way to realize this is that M-theory contains also membranes and these in general can wrap the compactified circle, in the boosted frame these membranes become non-local while the longitudinal modes remain local. From the type IIA point of view the only local elements are the D0-branes. Thus, this sector decouples from the rest of the system. The N -th Kaluza-Klein mode corresponds to N units of momentum $\frac{1}{R_s}$ in the compact dimension, and in this case, the $U(1)$ sector is the R-R one form of type IIA. Hence, we interpret D0-charge as eleven-dimensional momentum. Having sketched the connection of D0-branes with supergravitons (of eleven-dimensional M-theory), one might ask what about the rest of the objects like membrane or fivebrane? As it is nicely explained in [11], there is no need to add extra degrees of freedom in the action to describe also these objects, but rather they can be described via D0-branes and T-dualities.

The procedure of boosting, relates the non-relativistic ten dimensional theory of many D0-branes (large N) with a particular frame of the M-theory, the so-called infinite momentum frame (IMF). Then the conjecture is that the dynamics of the D0-branes are in one-to-one correspondence with the Hermitian matrices of the $(0+1)$ -dimensional Yang-Mills theory and they describe all the contents of M-theory in this particular IMF frame¹. A graphical representation of the above is given in figure 1. The previous statement holds when N is large. Further investigations have proposed that this should be true even at finite N [12]. However since the equivalence principle breaks down for the finite N version of the theory [13], [14] it is quite unlikely that it can be related to a smooth theory of Einstein-Hilbert gravity. On the other hand, this might not be a real problem since in the spirit of holography, finite N corrections seem to correspond to quantum corrections, and from many different approaches quantum spacetime is believed to be discrete. Thus, it is more likely that the finite N version of the model would correspond to a (quantum corrected) gravity theory on a non-commutative space.

Another realization of this duality which might be seen as evidence is the following. Firstly, there is lack of a concrete microscopic description of M-theory, and since the contents of M-theory are supermembranes, the UV completion of this theory has to be regularized via these membranes. This procedure [15] results in a Hamiltonian containing bosonic (X 's) and fermionic (Ψ 's) terms

$$H_0 = R \text{Tr} \left(\frac{1}{2} \dot{X}_A \dot{X}_A - \frac{1}{4} [X_A, X_B]^2 - \frac{1}{2} \Psi^T \gamma^A [X_a, \Psi] \right), \quad A, B = 1, \dots, 9. \quad (1.3)$$

In the same Hamiltonian, one can arrive by the dimensional reduction of the $U(N)$ 10D-SYM theory. Also, since the strong coupling limit of type IIA string theory is supposed to be M-theory, one should also arrive in (1.3) by using type IIA terminology. Indeed, this is the case for a large number of D0-branes at low energy and weak coupling [5] (BFSS matrix model), [16]. Thus, we see that one can reach this particular Hamiltonian (1.3) by three different approaches, M-theory, type IIA string theory and SYM theory, and this strongly suggests that there must be an underlying relation among them.

It was only later when another proposal [6] (BMN matrix model), realized also [5] as a special case of a one parameter (μ) family of maximally supersymmetric (32 supercharges) matrix models, whose

¹Note however that the M5-brane of M-theory can not be easily realized from the Matrix model [6].

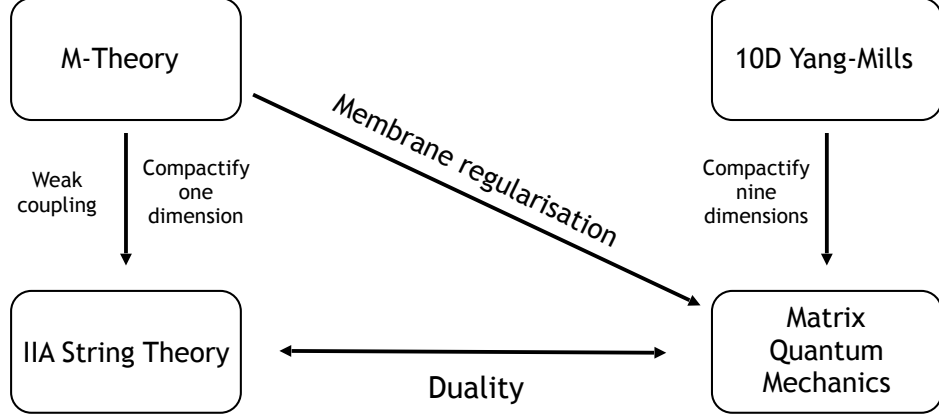


Figure 1: The correspondence between supersymmetric matrix quantum mechanics and M-theory. In this limit the type IIA string theory is in a specific frame, the infinite momentum frame ($N \rightarrow \infty$).

Hamiltonian is a deformation of (1.3) by

$$H = H_0 + \frac{R}{2} \text{Tr} \left(\sum_{i=1}^3 \left(\frac{\mu}{3R} \right)^2 X_i^2 + \sum_{a=4}^9 \left(\frac{\mu}{6R} \right)^2 X_a^2 + i \frac{2\mu}{3R} \epsilon^{ijk} X_i X_j X_k + i \frac{\mu}{4R} \Psi^T \gamma^{123} \Psi \right). \quad (1.4)$$

The original proposal [6] for general flux μ was supposed to give a description of the Discrete Light Cone Quantization (DLCQ) of M-theory on the maximally supersymmetric pp-wave background² of eleven dimensional supergravity

$$ds^2 = -2dx^+ dx^- + \sum_{A=1}^9 dx^A dx^A - \left(\sum_{i=1}^3 \frac{\mu^2}{9} x^i x^i + \sum_{a=4}^9 \frac{\mu^2}{36} x^a x^a \right) dx^+ dx^+, \quad (1.5)$$

with

$$F_{123+} = \mu \quad , \quad x^\pm = \frac{t \pm x}{\sqrt{2}}. \quad (1.6)$$

In the large N limit with fixed momentum $p = \frac{N}{R}$ and fixed μ , this matrix quantum mechanics model should describe M-theory on the pp-wave (1.5) background in a sector with boost-invariant momentum parameter μp .

²For general plane-wave backgrounds see [17].

We can write the bosonic part of the potential of (1.4) in sum of positive terms such as

$$V = \frac{R}{2} \text{Tr} \left[\left(\frac{\mu}{3R} X^i + i\epsilon_{ijk} X^j X^k \right)^2 + \left(\frac{\mu}{6R} \right)^2 X_a X^a \right]. \quad (1.7)$$

Then the types of vacua of this theory are manifest. They are

- $X^a = 0 = X^i$, $i = 1, 2, 3$, $a = 4, \dots, 6$ and corresponds to the trivial vacuum
- $X^a = 0$, $a = 4, \dots, 9$ and $X^i = \frac{\mu}{3R} J^i$, $i = 1, 2, 3$, with J^i being the generators of $SU(2)$ and corresponds to the non-trivial vacuum.

Among the quantum mechanics data and the gravity data there is only one relationship

$$\text{matrix elements } X \longleftrightarrow \text{D0-branes}, \quad (1.8)$$

and this is sufficient to reconstruct all the contents of gravity. The last statement has to do with the remarkable properties of D0-branes, which in presence of suitable fluxes they can polarize into D2- and NS5-branes. Then studying only X amounts to study all the contents of the gravity theory in the following way. Each diagonal element of X corresponds to the position of each D0-brane in the gravity side, and the non-diagonal elements to stringy interactions among the D0's, see figure 2. This is understood in the following way. Being the matrices X (sub)part of the Yang-Mills theory³ they also transform under the adjoint representation of $U(N)$ symmetry because of unitarity. We recall that under the same representation transforms an open string. Hence, both for the matrix and the string we need two indices to label the "state", which from the matrix point of view these indices can be realized as indices labelling the entries of the matrix, while from string perspective these are the Chan-Paton indices. Hence, for instance a string stretching from D-brane 1 to D-brane 2 is labelled as 12 and corresponds to the x_{12} entry of the matrix. Similarly, strings ending on the same D-brane are denoted with same indices, but since D0-branes are point-like the information we get is that this open string actually begins and ends on the same point. Then it is understood that since the size of the matrix is $N \times N$ we have to take N large such that we reconstruct more and more information about the gravity data.

Furthermore, the BMN matrix model has one more scale given by the mass μ . Then one can construct a dimensionless, effective coupling as $g_{\text{eff}} = \frac{\lambda}{\mu^3}$, where λ is the t'Hooft coupling $\lambda = g_{YM}^2 N$. This effective coupling controls the regimes of validity of the descriptions. Weak coupling ($g_{\text{eff}} \ll 1$) is when the matrix theory is valid and perturbation analysis can be applied with g_{eff} as expansion parameter. Strong coupling ($g_{\text{eff}} \gg 1$) is when the supergravity description is valid. In this paper we are interested in the gravity description for the bosonic sector of the model.

In addition we comment to the interpretation of the solutions of the BMN potential (1.7) from the gravity point of view. The trivial vacuum corresponds to placing N D0-branes in $r = 0$ in the gravity side creating in this way a black hole-like object in type IIA supergravity with $SO(3) \times SO(6)$ symmetry. In the BFSS model this realizes the metric of N coincident D0-branes since the $SO(8)$ symmetry naturally appears there. From M-theory point of view, this vacuum is believed to be due to a single 5-brane [18]. On the other hand, the number of non-trivial vacua can be organized in

³In particular it is dimensional reduction of it as we saw.

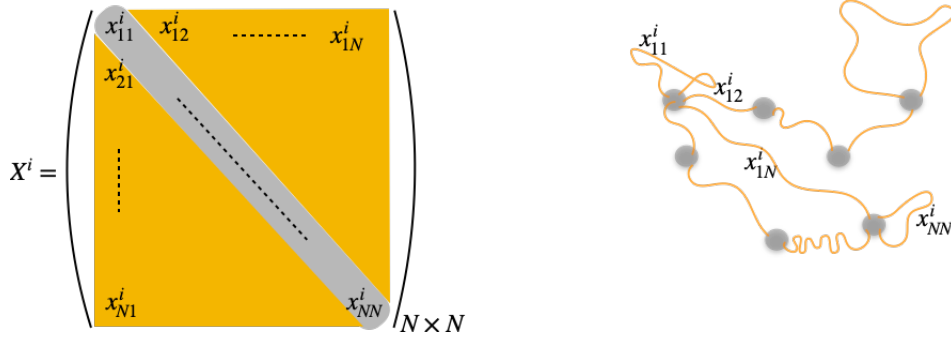


Figure 2: The correspondence between the matrices of quantum mechanics and D0-branes of string theory. Diagonal elements contain information for the positions of D0's while non-diagonal elements indicate interactions between them. The D0-branes are point-like, hence an entry $x_{\alpha\alpha}$ gives the "location" of the α -th D0-brane.

partitions of N as $\{N_1, \dots, N_k\}$. Geometrically they are supposed to correspond to concentric fuzzy shells in three dimensions placed at the center of the transverse six dimensions, with each shell having radius analogous to it's representation and the flux parameter μ , i.e $r_i \sim \frac{N_i \mu}{6R}$ [6],[19].

Nevertheless there has being a lot of progress towards understanding of the matrix models in the context of (quantum) gravity [20], [21], [22], [23], [24]. In this paper, we have organized the contents as follows. In the next section we investigate the dynamical behavior of D0-branes in a specific background, in particular in a symmetry reduced, classical pp-wave background. We do this by firstly calculating the force that a single (probe) D0-brane feels once it has been inserted in this background and secondly by calculating the force a probe D2-brane feels in the same background. One can think that this D2-brane can be realized as consisting of a large number ($q \gg 1$) of D0-branes taking into account a specific limit (see section 2.2). Furthermore, in section 3 we repeat the analysis for a D0-brane in a background that geometrically corresponds to the trivial vacuum solution of the BMN both in zero and finite temperature.

2 Force in symmetry reduced pp-waves: a warm up

In this section we start by calculating the force that a D0-brane feels in a classical plane wave background in ten dimensions and then repeating the calculation for a D2-brane. The starting point is an eleven dimensional metric which is a symmetry reduced version of the (1.5)

$$ds^2 = -(1 + \frac{Q^2}{4})dt^2 + (1 - \frac{Q^2}{4})(dx^{10})^2 - \frac{Q^2}{2}dtdx^{10} + \sum_{i=1}^3 dx_i dx_i + \sum_{a=4}^9 dx_a dx_a, \quad (2.1)$$

where Q is defined as ⁴

$$Q^2 \equiv \frac{\mu^2}{9} \sum_{i=1}^3 (X^i)^2 + \frac{\mu^2}{36} \sum_{a=4}^9 (X^a)^2, \quad (2.2)$$

and the 4-form field is given as

$$Q_{123(10)} = -Q_{0123} = \frac{\mu}{2}. \quad (2.3)$$

⁴This μ is the deformation parameter, and it is the same that appears in the matrix model (1.4).

We note that this metric, is written in the unboosted frame. Since any pp-wave metric is boost covariant [17], the effect of boosting is simply to rescale the deformation parameter μ . Furthermore, we note that this background is a general background that should capture general (but symmetry reduced) features of the BMN matrix model and in fact one can derive the action of the BMN from this metric [19], [16].

2.1 A D0-brane feels only the trivial vacua

We want to perform a compactification of (2.1) along the spacelike x^{10} direction to make contact with the ten-dimensional theory. One easy way to obtain this is to write down the relation between the 11D-metric and the 10D-metric

$$ds_{11}^2 = e^{-\frac{2\phi}{3}} ds_{10}^2 + e^{\frac{4\phi}{3}} (dx^{10} - C_\mu dx^\mu)^2. \quad (2.4)$$

Then, the ten-dimensional metric is given as

$$ds_{10}^2 = - \left(1 - \frac{Q^2}{4}\right)^{-\frac{1}{2}} dt^2 + \delta_{ab} \left(1 - \frac{Q^2}{4}\right)^{\frac{1}{2}} dx^a dx^b, \quad (2.5)$$

with

$$e^{\frac{4\phi}{3}} = 1 - \frac{Q^2}{4}, \quad C_0 = -\frac{Q^2}{4-Q^2} dt, \quad Q_{0123} = -\frac{\mu}{2}, \quad H_{123} = \frac{\mu}{2}. \quad (2.6)$$

Here C_0 is the RR-one form potential, H_{123} is the NS-NS three-form potential and F_{0123} is the RR three-form field strength. These form fields arise from the distribution of the eleven-dimensional three form into the R-R and NS-NS sector. Geometrically, this corresponds to distributing the eleven dimensional momentum into different geometric configurations of D-branes. When having in mind a single D0-brane we will consider only the one-form since it is the one that the D0-brane is naturally coupled with.

By direct calculation of the Ricci scalar of (2.5), we find

$$\begin{aligned} \mathcal{R} = \frac{\mu^2}{5184} \left(1 - \frac{Q^2}{4}\right)^{-\frac{5}{2}} \sum_{i=1}^3 \sum_{a=4}^9 \left\{ \mu^2(7 - Q^2) \left[(X^a)^4 + 48(X^a)^2(X^i + 3) + 96X^a(X^i)^2 \right. \right. \\ \left. \left. + 16(X^i)^2((X^i)^2 + 36) \right] \right. \\ \left. + 252[(X^a)^2 + 4((X^i)^2 + 9)]Q - 63[(X^a)^2 + 4((X^i)^2 + 9)]Q^3 \right\} \end{aligned} \quad (2.7)$$

From this we see that we can trust the supergravity solution (2.5) only in the low curvature region, i.e when $\frac{Q^2}{4} \ll 1$.

In this limit one can expand as follows

$$\begin{aligned} e^{-\frac{2\phi}{3}} &\approx 1 + \frac{Q^2}{8} + \mathcal{O}(Q^4) \\ C_0 &\approx -\frac{Q^2}{4} + \mathcal{O}(Q^4). \end{aligned}$$

Having set up the playground, now we turn to calculate the force. When we do this, we have in mind a situation when we naively place a D0-brane probe in the spacetime and ask for its dynamical behaviour. To this end, we need to write down the DBI action for the D0-brane by choosing the gauge where the parameter of the worldline of the D0 coincides with the time component of the spacetime. In a general background, it is given by

$$S_{D_0} = -T_0 \int dt e^{-\phi} \sqrt{-g_{\mu\nu} \frac{dx^\mu}{dt} \frac{dx^\nu}{dt}} + T_0 \int dt C_0, \quad (2.8)$$

where $T_0 = \frac{M_s}{g_s}$ is the tension of the D0-brane and the expression inside the square root is the pull-back of the spacetime metric on the worldline of the D0-brane. We remind that a Dp-brane naturally couples with a p+1 form, thus the only form that couples (naturally) to a single D0-brane is the one-form potential C_0 . Then, the action reads as

$$S_{D_0} = -T_0 \int dt \left(1 + \frac{3Q^2}{16} \right) \sqrt{1 + \frac{Q^2}{8} - \left(1 - \frac{Q^2}{8} \right) \dot{x}^2} + T_0 \int dt \left(-\frac{Q^2}{4} \right). \quad (2.9)$$

Moreover, after taking the non-relativistic limit we expand the action up $\dot{x}^2$ and Q^2 terms leading to

$$S_{D_0} \approx -T_0 \int dt \left[1 + \frac{Q^2}{2} - \frac{1}{2} \dot{x}^2 \right] + \mathcal{O}(Q^4, \dot{x}^4). \quad (2.10)$$

One can read off the potential from this action:

$$V_{D_0}(x) = T_0 \left(1 + \frac{Q^2}{2} \right). \quad (2.11)$$

Recalling (2.2) we write the potential as,

$$V_{D_0}(X^i, X^a) = T_0 \left[1 + \frac{1}{2} \left(\frac{\mu^2}{9} \sum_{i=1}^3 (X^i)^2 + \frac{\mu^2}{36} \sum_{a=4}^9 (X^a)^2 \right) \right]. \quad (2.12)$$

We see an explicit dependence of μ in the potential. It is also worth noting that the trivial vacua ($X^i = 0$ and $X^a = 0$) are supersymmetric solutions of the potential after performing a shift, but the non-trivial $SU(2)$ vacua are not. The latter, geometrically are believed to correspond to concentric fuzzy spheres, however the single D0-brane in this background is agnostic if there are any fuzzy spheres at all. Had we consider a different probe, for example a D2-brane, then this would couple to higher forms and would give rise to the $SU(2)$ part of the potential. This is demonstrated in the next subsection.

Furthermore, we can write the terms inside the sum as the radii of an S^2 and S^5 sphere. In other words we can write the potential as

$$V_{D_0}(r_2, r_5) = T_0 \left[1 + \frac{1}{2} \left(\frac{\mu^2}{9} r_2^2 + \frac{\mu^2}{36} r_5^2 \right) \right], \quad (2.13)$$

where r_2 denotes the radius of the S^2 sphere and r_5 the one of S^5 .

From this potential one can obtain the force that the D0-brane feels by differentiating with respect to the radial coordinates. This is simply given by

$$F_{D_0}(r_2, r_5) = -T_0 \frac{\mu^2}{9} \left(r_2 + \frac{r_5}{4} \right). \quad (2.14)$$

However, this is a naive picture because the hole system is studied at the classical level. According to matrix model simulations, the matrices are never zero but they subject to quantum fluctuations independent of which vacuum one studies [20], [21], [22], [24].

2.2 A D2-brane feels the $SU(2)$ vacua

In this section we calculate the force that a D2-brane probe feels in the weakly curved background (2.5). One can think this D2 probe as being made of (q) D0-branes where the number of D0's is taken to be large. However, in order to consider this as a probe, one has to require that the probe charge q is much less than the background charge N such that we do not take backreaction into account, i.e we are studying the limit $1 \ll q \ll N$. We can imagine this brane to have a S^2 topology embedded in a three dimensional space. In particular, one natural embedding would be along the $SO(3)$ part of the metric where the other six dimensions are transverse to it. For convenience, we place the D2-brane in the center of the the six transverse directions such that $X^a = 0$. Then the DBI and WZ action of the D2-brane will be of the familiar form

$$S_{D_2} = -T_2 \int d^3\sigma e^{-\phi} \sqrt{-\det(G_{ab} + 2\pi\alpha' \mathcal{F}_{ab})} + T_2 \int (C_3 + 2\pi\alpha' \mathcal{F}_2 \wedge C_1), \quad q \ll N \quad (2.15)$$

where $2\pi\alpha' \mathcal{F}_2 = 2\pi\alpha' F_2 - B_2$. As in the case of the D0-brane, we have also chosen the static gauge here, which corresponds to identify the worldvolume dimensions with the three dimensional part of the sub space of the ten-dimensional spacetime. We can choose t, θ, ϕ coordinates parametrizing the S^2 . The radius of S^2 is given by r , while the angles have the usual ranges $\theta \in [0, \pi]$, $\phi \in [0, 2\pi]$. For a compact D2-brane of arbitrary topology the total charge q is quantized and it is given as

$$\int_{S^2} F_2 = 2\pi q \quad \longrightarrow \quad F_2 = \frac{1}{2} q \sin \theta d\theta \wedge d\phi. \quad (2.16)$$

We note that the integer q is also related to the first Chern class of the $U(1)$ bundle at a fixed point in time [25], [26] and moreover, it acts as a source for the R-R vector field and the interpretation of it is the number of D0-branes that are bound on the surface of the D2-brane.

Analogously, the forms are given in the static gauge as

$$C_1 = \left(1 - \frac{Q^2}{4}\right)^{-1} dt \quad (2.17)$$

$$C_3 = \frac{1}{3} \mu dt \wedge S_2, \quad S_2 = \frac{1}{2} \epsilon_{ijk} x^i \wedge dx^j \wedge dx^k, \quad i, j, k = 1, 2, 3. \quad (2.18)$$

Furthermore, the pullbacks of the metric G_{ab} on the D2-brane are given as

$$\det g_{tt} = - \left(1 - \frac{Q^2}{4}\right)^{-\frac{1}{2}} \quad (2.19)$$

$$\det g_{\perp} = \det(g_{\theta\theta} g_{\phi\phi}) = \left(1 - \frac{Q^2}{4}\right) r^4 \sin^2 \theta. \quad (2.20)$$

From these we have that

$$F_{\theta\phi} = \frac{1}{2}q \sin \theta \quad (2.21)$$

$$F_{ab}F^{ab} = \frac{q^2}{2\left(1 - \frac{Q^2}{4}\right)r^4} \quad (2.22)$$

$$\det F_2 = \frac{1}{2}F_{ab}F^{ab} \det g_{\perp} = \frac{q^2}{4} \sin^2 \theta. \quad (2.23)$$

The DBI action can be written as

$$S_{DBI} = -T_2 \int d^3\sigma e^{-\phi} \sqrt{-\det g_{tt}} (\det g_{\perp} + 2\pi\alpha' \det F_2)^{\frac{1}{2}}, \quad (2.24)$$

and considering the main contribution coming from $\det F_2$ ⁵, we can expand the action around this and get

$$S_{DBI} \approx -T_2 \int d^3\sigma e^{-\phi} \sqrt{-\det g_{tt}} \left(2\pi\alpha' \sqrt{\det F_2} + \frac{\det g_{\perp}}{4\pi\alpha' \sqrt{\det F_2}} \right). \quad (2.25)$$

Using the exact expressions this gives

$$\begin{aligned} S_{DBI} &\approx -T_2 \int dt d\theta d\phi \left[\left(1 - \frac{Q^2}{4}\right)^{-1} \pi\alpha' q \sin \theta + \frac{r^4 \sin \theta}{2\pi\alpha' q} \right] \\ &\approx -T_2 \int dt \left[\left(1 - \frac{Q^2}{4}\right)^{-1} 4\pi^2 \alpha' q + \frac{2r^4}{\alpha' q} \right]. \end{aligned} \quad (2.26)$$

Similarly the WZ action gives

$$S_{WZ} = T_2 \int dt \left[\frac{4\pi\mu r^3}{3} + 4\pi^2 \alpha' q \left(1 - \frac{Q^2}{4}\right)^{-1} \right]. \quad (2.27)$$

We see that the leading terms cancel each other, the subleading terms that remain can be mapped to the commutator term $\sim \text{Tr} [X^i, X^j]^2$ and the Myers term $\sim -\text{Tr} (i\frac{\mu}{3}\epsilon_{ijk}X^iX^jX^k)$ in the BMN model. In addition, one could also be interested in the contributions coming from the B_2 field. However, as it is shown in [27] the contributions of B_2 coming from WZ and DBI action cancel each other. Nevertheless, we can add a term $\sim \mu^2 r^2$ such that we complete the square in the action. This will make the interpretation more clear. By adding and subtracting this term, we write the action as

$$S_{D_2} \approx -\frac{2T_2}{\alpha' q} \int dt \left[\left(r^2 - \frac{2\pi\mu\alpha' q}{3} r \right)^2 - \frac{4\pi^2 \mu^2 \alpha'^2 q^2 r^2}{9} \right]. \quad (2.28)$$

The potential is simply

$$V_{D_2}(r) = \frac{2T_2}{\alpha' q} \left[\left(r^2 - \frac{2\pi\mu\alpha' q}{3} r \right)^2 - \frac{4\pi^2 \mu^2 \alpha'^2 q^2 r^2}{9} \right]. \quad (2.29)$$

⁵This amounts to $\frac{q^2}{4} \sin^2 \theta \gg 1$, reflecting the fact that the charge q is large but nevertheless we are in the limit $q \ll N$

From this we see that apart from the trivial $r = 0$ solution, there is a non-trivial equilibrium radius at

$$r = \frac{4\pi\alpha'q\mu}{3}. \quad (2.30)$$

This is understood as the radius in which the contraction that is caused by the r^2 and r^4 terms compensates with the expansion caused by the r^3 term in the potential. In fact one can generalize this and have a number (k) of concentric D2-branes instead of just one. Then for each D2-brane there is an equilibrium radius given by

$$r_i = \frac{4\pi\alpha'q_i\mu}{3}, \quad i = 1, \dots, k, \quad (2.31)$$

with q_i being the charge of the i -th D2-brane. In the Matrix model side instead, this picture corresponds to a particular copy of the N dimensional representation of $SU(2)$ up to a non-commutative factor in the case of a single D2-brane and a partition $\{N_1, \dots, N_k\}$ of the N dimensional representation of $SU(2)$ in the case of k concentric D2-branes.

The force that the D2 feels from the background is given as

$$F_{D_2}(r) = -\frac{8T_2}{\alpha'q}r^2(r - \pi q\alpha'\mu). \quad (2.32)$$

In fact, we see that it is independent of the configuration of the geometry, for example independent of the warp factor $e^{\frac{2\phi}{3}}$, but it does depend on its own charge q and the flux parameter μ , in accordance with the D0-brane case. We also see that the force becomes repulsive for $r \in (0, \pi q\alpha'\mu)$. This results to an interpolation from a strong, attractive force to a slightly repulsive force after going through the minimum of the potential. In the minimum, we have a D2-brane freely sitting on a position in spacetime. So in this way, we can see that we probe and reveal different parts of the geometry since this picture in the D0-probe was absent.

3 BMN vacua

In this section, after discussing the matrix model vacua and how one can think this as a electrostatic configuration problem following [28], we describe the gravity duals and we compute the force of the D0-brane for the trivial vacuum of the BMN. For similar discussions see [29], [30], [31].

3.1 Generic analysis

In general, we should look for a gravity solution that preserves half of the supersymmetries⁶ and respects the $SO(3) \times SO(6)$ symmetry. This can be motivated from the specific form of the Hamiltonian of the matrix model (1.4) since the gravity solution should respect the symmetries therein. The most general solution in type IIA string theory that meets these conditions is [28]

$$ds_{10}^2 = \left(\frac{\ddot{V} - 2\dot{V}}{-V''} \right)^{\frac{1}{2}} \left[-\frac{4\ddot{V}}{\ddot{V} - 2\dot{V}} dt^2 - \frac{2V''}{\dot{V}} (d\rho^2 + dz^2) + 4d\Omega_5^2 + 2\frac{V''\dot{V}}{\Delta} d\Omega_2^2 \right], \quad (3.1)$$

⁶This is because the BMN model has various BPS states, but here in particular we are interested for 1/2-BPS states.

with

$$\begin{aligned}\dot{V} &= \rho \partial_\rho V, \quad \ddot{V} = \rho \partial_\rho (\dot{V}), \quad V' = \partial_z V, \quad V'' = \partial_z^2 V, \\ \Delta &= \left(\ddot{V} - 2\dot{V} \right) V'' - \left(\dot{V}' \right)^2,\end{aligned}$$

and the dilaton and NS-NS and R-R forms are given as

$$\begin{aligned}e^{4\phi} &= \frac{4(2\dot{V}-\ddot{V})^3}{V''V^2\Delta^2}, \quad B_2 = 2 \left(\frac{\dot{V}\dot{V}'}{\Delta} + \rho \right) d\Omega_2 \\ A_1 &= \frac{2\dot{V}\dot{V}'}{2\dot{V}-\ddot{V}} dt, \quad A_3 = -\frac{4\dot{V}^2 V''}{\Delta} dt \wedge d\Omega_2.\end{aligned}\tag{3.2}$$

In (3.1) appears a function $V(z, \rho)$. In fact this function can be interpreted as a potential that sources different brane (NS5, D2, D0) configurations, and depending on the specific form of this function one realizes different gravity solutions, which correspond to different vacua in the matrix model side. We are interested in the bosonic sector of BMN because in that case we can write a metric ansatz that respects the bosonic symmetries. In the BMN case, we see from (1.4) that these symmetries are $R \times SO(3) \times SO(6)$. The first of these implies that there is a Killing vector associated to translations of the time coordinate. The second, implies that we can imagine a sector of the spacetime to consist of a S^2 sphere⁷ while the third the existence of a S^5 sphere, on which the bosonic generators act. Thus one can analyse solutions in eleven dimensions by using only three variables, say x_1, x_2, y which satisfy the Toda equation in three dimensions [32],[28]

$$(\partial_{x_1}^2 + \partial_{x_2}^2) V + \partial_y^2 V = 0.\tag{3.3}$$

In fact, it turns out that $y = r_2 r_5^2$, where r_2, r_5 are the radii of the S^2 and S^5 respectively. Exploiting the translational invariance under one coordinate (x_1), and after changing variables $(x_2, y) \mapsto (\rho, z)$ one can recast this equation to the Laplace equation in a three dimensional space [32]

$$\frac{1}{\rho} \partial_\rho (\rho \partial_\rho V) + \partial_z^2 V = 0.\tag{3.4}$$

We note that this three-dimensional space is not a sub-part of the ten dimensional one but rather maps to an electrostatic problem in a three dimensional space, where all gravity solutions are classified by ρ, z . Here, both ρ and z are radial (scale) coordinates, the former captures the size of fivebranes and the latter the positions of the conducting disks, which is actually being set by the five-branes. If one, naively tries to find a gravity solution by just plugging in a solution of (3.4) into the metric (3.1) will end up with singular gravity solutions. To avoid this, one has to impose boundary conditions at $y = 0$ such that $V(y = 0) = 0$. This condition corresponds to regimes with vanishing radii of S^2 or S^5 . Indeed S^2 shrinks for constant z in the z, ρ space (see figure 3) and S^5 shrinks between two nearby disks in the $\rho = 0$ line.

In the case of the BMN model, the asymptotic expansion is decomposed in two parts of which the first one represents the electrostatic potential and the second one the specific electrostatic configuration imposed by the boundary conditions. We note that this means that one can specify the solution but not

⁷Actually it is more appropriate to say that we have non-commutative sphere since the coordinates that are assigned to $SU(2)$ generators do not commute.

necessary to which vacuum corresponds, since there will be vacua with the same boundary conditions. The first non-trivial potential is [18]

$$V \approx \rho^2 z - \frac{2}{3} z^3 + \mathcal{P} \frac{z}{(\rho^2 + z^2)^{\frac{3}{2}}}, \quad (3.5)$$

where $\mathcal{P}$ is a short of dipole sub-leading correction, produced by the conducting disks and their images with respect to the infinite conducting plane at $z = 0$. Its value is given by

$$\mathcal{P} = 2 \sum_i z_i Q_i \sim N, \quad (3.6)$$

where i runs over the various disks, Q_i is the charge of the i -th disk and N is the total number of D0-branes.⁸ The number of D0-branes can be extracted from the particular configurations of the disks. Suppose for example that there is a finite set of conducting disks at positions $z_i = \frac{\pi}{2} N_5^i$, with radii R_i and charge $Q_i = \frac{\pi^2}{8} N_2^i$. Then the disks are sourced by the NS5-branes in the background, with the charge Q_i being associated to the D2-brane charges. In this sense, one can write the number of D0-branes as

$$N = \sum_i \left(\sum_{j < i} N_5^j \right) N_2^i. \quad (3.7)$$

Furthermore, N_2^j is related to the copies of dimension $N_5^j = (2j+1)$ irreducible representations, giving a geometric realization of the situation of the matrix model.

Plugging (3.5) into the metric (3.1) and expanding at large ρ and z , the metric takes the form

$$ds^2 \approx -\frac{8z^{\frac{7}{2}}}{\sqrt{15\mathcal{P}}} dt^2 + 2\sqrt{15\mathcal{P}} z^{-\frac{7}{2}} (dz^2 + z^2 d\Omega_2^2 + d\rho^2 + \rho^2 d\Omega_5^2) + \dots, \quad (3.8)$$

where we have keep only the leading terms on z and ρ and the relevant piece of metric that still has the $SO(3) \times SO(6)$ symmetry. The dots above denote mixing and subleading terms. We note that the part inside the parenthesis is actually $\mathbb{R}^9$, which is written in a way that makes the symmetries of the theory manifest. One can further change coordinates

$$r^2 = 16 (\rho^2 + z^2), \quad (3.9)$$

and in the large r limit the metric (3.8) approximates the near horizon geometry of N coincident D0-branes

$$ds^2 \approx -H(r)^{-\frac{1}{2}} dt^2 + H(r)^{\frac{1}{2}} (dr^2 + r^2 d\Omega_8^2) + \dots, \quad H(r) \approx \frac{256 \cdot 15\mathcal{P}}{r^7}. \quad (3.10)$$

However the large r region is a bit subtle. Let us try to clarify some possible issues. The supergravity description is valid when $g_{\text{eff}} = \frac{\lambda}{\mu^3} \gg 1$ while for $g_{\text{eff}} \ll 1$ we can trust perturbative calculations in matrix model. The first case corresponds to a region between the black hole (i.e near the core of the geometry in figure 4) and $r \sim \lambda^{1/3}$, while the second case is near the boundary. However, near the boundary the gravity solution is not valid since the curvature is high (see next subsection). Hence we are in a region when r is large but nevertheless not near the boundary. It is

⁸Here we make contact with the introductory remarks, where we stated that all the contents can be reconstructed from D0-branes.

large with respect of the core of the geometry and a good estimate would be $r \lesssim \lambda^{1/3}$. Here we can imagine the following situation, we suppose that we want to study the system (for now) in a specific energy scale (i.e fixed λ), where we can trust the supergravity solution. This amounts to

$$r \gg 1 \quad \text{and} \quad g_{\text{eff}} = \frac{\lambda}{\mu^3} \gg 1, \quad \lambda = \text{fixed}, \Rightarrow \mu \rightarrow 0. \quad (3.11)$$

Thus, in this case and to leading order we recover the case of BFSS model and this is what the metric (3.10) tells us about.

3.2 Trivial vacuum

From the gauge theory side, it seems that the easiest classical vacuum, is the one that all matrices vanish trivially, i.e $X^i = 0 = X^a$, with $i = 1, 2, 3$, $a = 4, \dots, 9$. This corresponds to the situation where we have N copies of the trivial ($j = 0$) representation of $SU(2)$, and in the electrostatic description this translates to $N_2 = N$ and $N_5 = 1$, i.e a single disk at a distance close to (of order one unit) the infinite conducting disk at $z = 0$ (see figure 3). In the gravity side the picture is the following: N corresponds also to the number of D0-branes and putting these in the $r = 0$ position of the spacetime creates a singular geometry (3.10). However, the situation in BMN model is a bit different, because one has to realize also the correct symmetries, namely $SO(3) \times SO(6)$. This vacuum is believed to correspond to a single fivebrane in the gravity side [18], but the gravitational approximation is not very good near the fivebrane [6], [28]. Nevertheless, we perform an expansion in the metric in the large ρ limit while keeping z of order one. This translates to

$$r^2 = 16(\rho^2 + z^2) \gg 1, \quad \rho \gg 1, \quad z \sim \mathcal{O}(1). \quad (3.12)$$

3.2.1 Zero temperature

We will start applying this to zero temperature first and then move to finite temperatures in the next subsection. After taking the large r and then the trivial vacuum limit, we get a metric of the form

$$ds^2 = -H(r)^{-\frac{1}{2}} dt^2 + 16H(r)^{\frac{1}{2}} (dz^2 + z^2 d\Omega_2^2) + H(r)^{\frac{1}{2}} (dr^2 + r^2 d\Omega_5^2), \quad (3.13)$$

$$H(r) \equiv \frac{256 \cdot 15\mathcal{P}}{r^7}, \quad (3.14)$$

$$e^\phi = \left(\frac{2^{12} 15\mathcal{P}}{r^7} \right)^{\frac{3}{4}}, \quad (3.15)$$

$$A_1(r) = -\frac{r^7}{2^{11} 15\mathcal{P}} - \frac{7\pi^2 r^5}{4^5 15\mathcal{P}} - \frac{7\pi^4 r^3}{4^3 48\mathcal{P}} + \frac{r^2}{10} - \frac{7\pi^6 r}{4 \cdot 384\mathcal{P}} - \frac{3\pi^2}{5}, \quad (3.16)$$

We have kept only leading orders both in the expression of dilaton and the one form. The next correction to the dilaton is of order $\mathcal{O}(r^{-29/4})$ and in the large r limit we neglect it. In the same limit, the contribution of the R-R one form is significant. Furthermore, we see that in this limit and for same energy scale (or the same radius r), the effective curvature of the five-sphere, which in string units is the inverse of $\frac{\mathcal{R}_5^2}{\alpha'} = H(r)^{\frac{1}{2}} r^2$, is much smaller than the one of the two-sphere, which is the inverse of $\frac{\mathcal{R}_2^2}{\alpha'} = H(r)^{\frac{1}{2}} z^2$. This means that curvature corrections coming from the two-sphere part of the metric may become important when compared with the one of the five-sphere. In fact to make

the metric physically viable we should take the limit $z \sim \lambda^{1/3}$ [28]⁹. This amounts to blow up the S^2 part. We also note that the validity of the supergravity solution is when the dilaton is not too large and when the effective curvature of the five-sphere is not too big. Recalling $\mathcal{P} \sim N$ this translates to the range

$$N^{\frac{1}{7}} \ll r \ll N^{\frac{1}{3}}. \quad (3.17)$$

The Ricci scalar of (3.13), can be computed and it is given by

$$\mathcal{R} = -\frac{49\sqrt{\frac{3}{5}}}{64} \left(\frac{r^3}{\mathcal{P}}\right)^{\frac{1}{2}}. \quad (3.18)$$

We see that in the regime where the supergravity description for the trivial vacuum is a good approximation, the spacetime is almost flat provided that N is large.

Lastly, before moving to calculation of the force, it worths mentioning that after a coordinate rescaling and transformation, namely

$$z \rightarrow \frac{\zeta}{4}, \quad R \rightarrow \frac{2}{5} \frac{r_0^{5/2}}{r^{5/2}}, \quad r_0^7 \equiv 256 \cdot 15\mathcal{P}, \quad t \rightarrow r_0 t \quad (3.19)$$

the metric becomes

$$ds^2 = \left(\frac{2}{5}\right)^{7/5} \frac{r_0^2}{R^{7/5}} \left[-dt^2 + dR^2 + \left(\frac{5}{2}\right)^2 R^2 d\Omega_5^2 \right] + \left(\frac{2}{5}\right)^{-7/5} R^{7/5} (d\zeta^2 + \zeta^2 d\Omega_2^2). \quad (3.20)$$

We see that the near horizon limit of the vacuum solution is conformally equivalent to $AdS_2 \times S^5 \times B^3$ if we keep r_0 fixed which means to keep $\mathcal{P}$ fixed. We also note that the "size" of the $AdS_2 \times S^5$ part of the metric is much bigger than that of the B_3 part. Of course this was expected from the beginning since the trivial vacuum limit corresponds to large $r^2 = 16(\rho^2 + z^2)$, with ρ large and z of order one.

3.2.2 Force at zero temperature

In this subsection we calculate the force that a D0-brane feels in the background of (3.13) which corresponds to the trivial vacuum in the matrix model side. One might worry that since we are at zero temperature the force should be zero due to supersymmetry. However, we remind that so far we are interested in and analysing only the bosonic sector of the model and this is the force due to the bosonic degrees of freedom. We actually repeat the calculation of section 2. To this end we use (2.8) with the data of (3.13), fixing the spherical part of the metric. Doing this, and in the non-relativistic limit, we get the action

$$S_{D_0} \approx -T_0 \int dt \left[e^{-\phi} H(r)^{-\frac{1}{4}} \left(1 - \frac{1}{2} H(r) \left(\left(\frac{dz}{dt} \right)^2 + \left(\frac{dr}{dt} \right)^2 \right) \right) - A_1(r) \right] \quad (3.21)$$

From this DBI action, the potential can be read off as

$$V_{D_0}(r) = T_0 \left[e^{-\phi} H(r)^{-\frac{1}{4}} - A_1(r) \right]. \quad (3.22)$$

⁹Both r and z here are energy scales, and this condition reflects the fact that we push the regime of validity of the metric to its boundary. In [33], and based on [34] was argued that the breaking of Einstein supergravity starts at scales $\lambda^{1/3}$.

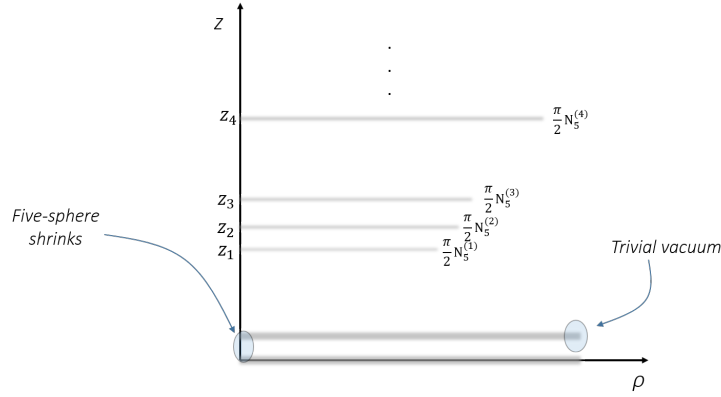


Figure 3: A cartoon of the $z-\rho$ plane. Constant z lines indicate that the two-sphere shrinks while the five-sphere shrinks in the region between nearby disks near $\rho = 0$. We have a conducting disk at $z = 0$ which extends to infinity in the ρ direction.

We remind that (3.13) corresponds to a deformed version of the near horizon geometry of D0-branes. These D0-branes are in fact coincident and realize a $U(N)$ symmetry. When we treat one of them as a probe, then we actually break the symmetry to $U(N) \rightarrow U(N-1) \times U(1)$, giving a Higgs expectation value¹⁰ r . This corresponds to placing a single D0-brane at distance r from the center.

We see the behavior of the force and the potential in figure 5. In fact we see an interesting behavior. Coming from large values of r towards $r = 0$ the D0-brane experiences different situations depending on r , i.e the distance of the probe from the bunch. Where, by the term bunch we are referring to the constituents of the black hole-like solution (3.13) which in this case is a bunch of D0-branes. To make the point more clear, suppose that we fix N to be very large $N \gg 1$ and vary r , keeping in mind the conditions (3.17). Indeed for large r , we see an attractive force which going towards the black hole-like core¹¹ becomes repulsive. We give a more general comment on this in the next section A.1, but the behaviour follows a pattern that can be described as follows: for sufficient large N the probe will experience an attractive force far away from the core, but going towards the core will feel repulsive phenomena. We remind that the whole analysis is classical and is valid in the regime (3.17). Nevertheless, at least *qualitatively* this is in accordance with [35] where the same effect was observed in quantum corrected black 0-brane geometries and it was verified by simulations in [24]. Also, it deserves to be mentioned that the steep curves of the potential and the force should not taken too seriously because they are outside the regime of validity (3.17) of the classical supergravity geometry (3.13). However, the behaviour seems to be independent of N since the curves join going towards the core.

What is interesting here in contrast to the BFSS case, is the fact that repulsive phenomena can occur even classically and far away from the horizon if N is sufficiently large. This is also the regime where the gravity description is valid, and we can understand this repulsive phenomenon as the R-R charge of the D0-branes.

Physically, it is also interesting the fact that in this system if a D0-brane is heading towards the

¹⁰We remind that we have set $\alpha' = 1$, otherwise the Higgs expectation value would be $U = \frac{r}{\alpha'}$.

¹¹Since we are at zero temperature, the horizon is located at $r = 0$, so approaching the horizon means approaching the singularity here.

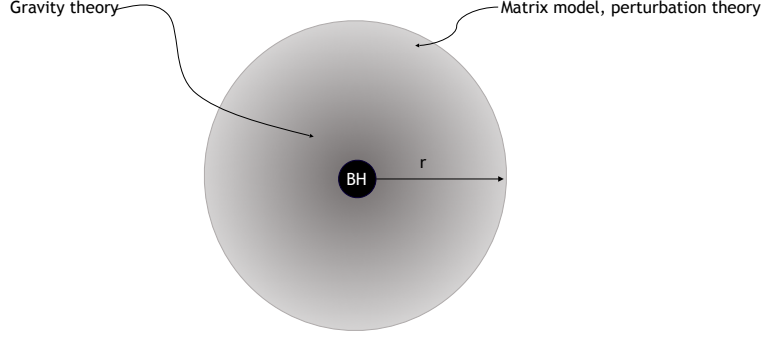


Figure 4: A pictorial representation of the regime of gravity and the regime of matrix model. In this way one can approximate different regimes with the help of the radial parameter r , the deformation parameter μ and the effective coupling $g_{\text{eff}} = \frac{\lambda}{\mu^3}$.

bunch when N is large, then it should have enough energy such that it will be able to overcome the potential which grows going towards the horizon and eventually merge with the bunch. One more observation is the fact that for different N the minimum of the potential varies with r . In fact one can imagine that instead of just one D0-brane for probe we have many of them ($q \ll N$)¹², then these D0-branes could be in the minimum of the potential on a specific distance r or in a specific energy scale of the system, forming in this way a bound state with the bunch.

It is known for the BMN model that there might be a tunneling effect between "nearby" vacua [36], where the probability to tunnel is given by the exponential $e^{-\frac{N}{\lambda}}$ since it is mediated by instantons. N is the number of the degrees of freedom (i.e D0-branes) and λ the t'Hooft coupling. As a result in the regime we are interested in, the tunneling between different vacua is exponentially suppressed.

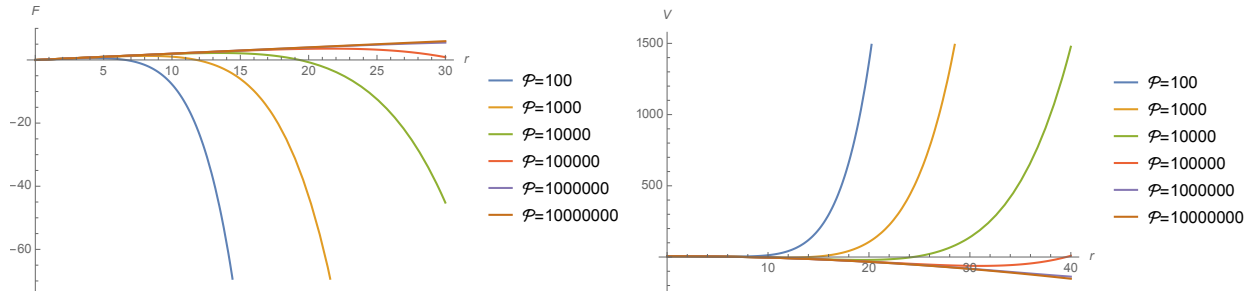


Figure 5: The force (left) and potential (right) are plotted with respect to the radial coordinate r for various values of $\mathcal{P} \sim N$.

¹²Ignoring the backreaction.

3.2.3 Force at finite temperature

At finite temperature the BMN model has 1/2- BPS states [6], [19] and we should expect the force to be present even if we consider fermions. Eventually, also we would like to make some contact with the quantum mechanics side. Note that this quantum mechanics however is a subsector of the SYM theory. So let us be a bit more illustrative here. The parameters that control the vacua of the BMN, are the number¹³ of D2-branes N_2 , the number of D5-branes N_5 , and the number of D0-branes N_0 , whose relation is given as [28]

$$N_2 = \frac{N_0}{N_5}. \quad (3.23)$$

Now, we remind that the trivial vacuum is dictated by $N_5 = 1, N_2 = N_0 = N$, with N being the copies of the trivial ($j = 0$) representation. This particular configuration is given in figure 3. Then to understand the situation in the gauge theory side, we need to convert the gravity data to the gauge data. By direct comparison with [37], we find the following [28]¹⁴

$$\rho_0^4 = \frac{\pi^4}{2} g_{YM}^2 N, \quad d = \frac{\pi}{2} N_5 = \frac{\pi}{2}, \quad \mathcal{Q} = \frac{\rho_0^4}{8}, \quad \mathcal{P} = 2d \mathcal{Q} = \frac{\pi^5 g_{YM}^2 N}{16}. \quad (3.24)$$

Here, ρ_0 is the value of the coordinate ρ on which the tip of the conducting disk lies (see figure 3), d is the charge of N_5 which is also the distance between the two conducting disks. Before we continue, the above data reflect the repeated statement, that the size of the fivesphere is much larger than the one of S^2 , into the radius of the fivesphere [28]

$$\frac{R_{S^5}^2}{\alpha'} = 4\pi\rho_0 = 4\pi \left(\frac{\lambda}{2\mu^3} \right)^{\frac{1}{4}}. \quad (3.25)$$

We investigate the gravity solution at finite temperature T in this vacuum. We can write the finite T black brane solution as

$$\begin{aligned} \frac{ds_T^2}{\alpha'} &= -H(r)^{-\frac{1}{2}} f(r) dt^2 + 16H(r)^{\frac{1}{2}} (dz^2 + z^2 d\Omega_2^2) + H(r)^{\frac{1}{2}} \left(\frac{dr^2}{f(r)} + r^2 d\Omega_5^2 \right), \\ H(r) &= \frac{\lambda d_0}{r^7}, \quad e^\phi = 2^6 \left(\frac{\lambda 15 \pi^5}{r^7} \right)^{\frac{3}{4}}, \quad f(r) = 1 - \frac{r_0^7}{r^7}, \\ A_1(r) &= -\frac{r^7}{1920\pi^5 \lambda} - \frac{7r^5}{960\pi^3 \lambda} - \frac{7r^3}{192\pi \lambda} + \frac{r^2}{10} - \frac{7\pi r}{96\lambda} - \frac{3\pi^2}{5}, \\ d_0 &\equiv 240\pi^5, \quad \lambda \equiv g_{YM}^2 N, \quad g_s = 4\pi^2 l_s^3 \frac{\lambda}{N}, \quad G_{10} = g_s^2 l_s^8, \end{aligned} \quad (3.26)$$

and the horizon is located at $r = r_0$, with r_0 representing the thermal energy scale. This is useful to study thermodynamic properties of the system. Note that the horizon is dictated only from the radius of the five-sphere, possibly reflecting the fact that this particular vacuum solution is due to a single fivebrane. Indeed from this metric one can extract the temperature via the surface gravity

$$T = \frac{7r_0^{5/2}}{4\pi\sqrt{\lambda d_0}} \quad (3.27)$$

¹³To be more precise it is the charge of the various D-branes.

¹⁴We suppress μ dependence here by setting $\mu = 1$ but we can restore it by using $g_{\text{eff}} = \frac{\lambda}{\mu^3}$ and by dimensional analysis, e.g $[\alpha'] = 2, [\mu] = 1, [\lambda] = 3$ in units of energy.

but before we continue let us state when the gravity description is valid. In principle, the gravity solution due to the fivebrane is not well behaved and this is reflected from the fact that if z is of order one, then the curvature of S^2 gives a different limit of validity. To fix this, we demand $z \gg 1$ [28] such that the metric is well behaved. Thus, the curvature of the spheres is not too large from which we get the conditions

$$r \lesssim \lambda^{1/3} \quad \text{from the } S^5, \quad (3.28)$$

$$r \lesssim (\lambda z^4)^{1/7} \quad \text{from the } S^2, \quad (3.29)$$

where $z \sim \lambda^{1/3}$. In fact, we have shifted the problem to the size of the spheres, because sending $z \rightarrow \lambda^{1/3}$ means blowing up the two-sphere. However, this is the "extremal" case when z and r are of the same order and the sizes of the spheres are equal, but in general we have $z \ll r$. There is also one more constraint coming by demanding that the dilaton is not too large on the horizon which results in

$$\lambda \beta^3 \lesssim N^{10/7}, \quad (3.30)$$

and for sufficiently large N with respect to the combination $\lambda \beta^3$ the last condition is automatically satisfied.

The calculation of the force uses a variation of equation (3.22),

$$V_{D_0}(r) = T_0 \left[e^{-\phi} H(r)^{-\frac{1}{4}} f(r)^{\frac{1}{2}} - A_1(r) \right]. \quad (3.31)$$

with the data of the finite temperature gravity solution (3.26). We expect qualitatively similar behaviour as in the zero temperature system because of BPS states, and evaluating indeed we see the behavior in figure 6. Here we show the dependence of everything in terms of λ to make contact with the existing literature. One, can see that there is a μ dependence also if we use the dimensionless coupling $g_{\text{eff}} = \frac{\lambda}{\mu^3}$.

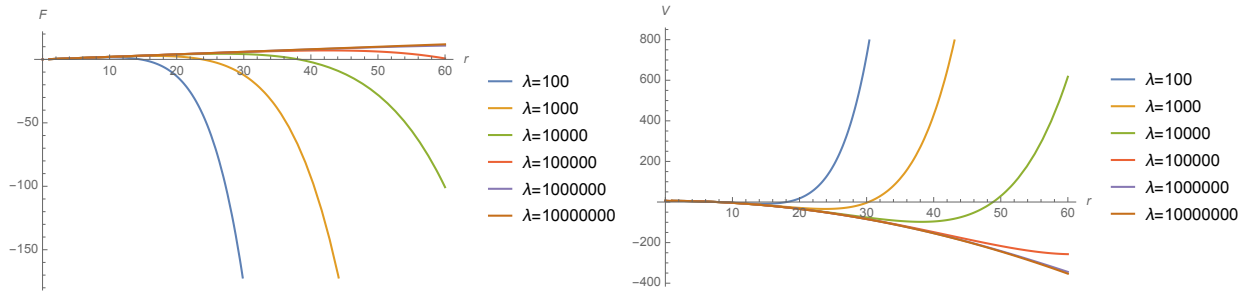


Figure 6: Finite temperature force (left) and potential (right) with respect to the radial coordinate r (or equivalently temperature T). We observe a similar behavior as in figure 5. We have set for the horizon $r_0 = 1$ and $g_{YM} = 1$.

We also note that the geometry which is believed to correspond to BFSS model is recovered when we take $\mu \rightarrow 0$. At first sight this might not be obvious but we observe the following. We always want the gravity description to be valid which results in $g_{\text{eff}} \gg 1$ and we also want $r \gg 1$ such that we are away from the singularity. If we fix λ which amounts to fix the energy scale of the system, we have that

$$g_{\text{eff}} = \frac{\lambda}{\mu^3} \gg 1, \quad \lambda = \text{fixed} \Rightarrow \mu \rightarrow 0. \quad (3.32)$$

Then for fixed $\lambda \gg \mu^3$ with $\mu \rightarrow 0$, the dominant contribution in the R-R one-form comes from the r^7 term for large r . This is also the case for BFSS model where the R-R form is given in terms of r^7 [33], [37]. This limit also is what constructs the metric (3.10) to leading order, which is the metric for the gravity dual of the BFSS.

4 Synopsis

We have demonstrated gravitational behaviors of D-brane probes (D0-and D2-branes) for symmetry reduced pp-wave backgrounds. In particular for the D0-probe we have also studied its behavior for geometry which corresponds to the trivial vacuum of BMN. We have seen a repulsive behavior as we approach the "naked" singularity in the extremal case and the horizon in the non-extremal case. This most likely indicates that there is a mechanism that gives rise to repulsive forces as one approaches the core. We have demonstrated, that this mechanism can occur both at classical level and also far away from the core for the BMN model. One possible way to understand this intuitively is because as an observer (here a D0-brane) approaches the core, firstly meets the S^5 part of the horizon surrounding the core and then the B^3 ball. This particular probe does not "see" the $SU(2)$ part of the geometry since in the potential does not appear the scale z that controls the size of B^3 . This is a choice based on observations made in subsection 2.1. There we have showed that a single D0-brane is agnostic for the existence of $SU(2)$ while in the next subsection we demonstrate that a D2-brane is aware. Now the charges of B^3 and S^5 consisting from D0-branes carry one forms with which the probe brane couples naturally. However, having the same charge the one form $A_1(r)$ causes a repulsion. Moreover, this is something that classically does not appear in the BFSS model but rather when one considers quantum corrections [35].

Nevertheless, the regime where we can trust supergravity depends on N and hence for large N one is already far away from the singularity. In the same regime, dynamically it seems more preferable that the D-brane probe will be at the minimum of the potential for specific N forming in this way a bound state with the bunch. This follows from the analysis done in the main text and from the principle of least action. From the same principle and figure 6, we conclude that if one perturbs the (D0-brane) probe energetically is more likely that it will head towards the bunch. This analysis holds only in the trivial vacuum geometry, and further studies need to be carried out to understand the situation of the non-trivial vacua which from geometric point of view are significantly more complicated. The limit which we took to construct this geometry is also realized in simulations (Cite to be appeared?). We are referring to the large deformation of the geometry, as in the main text several times was emphasized that the horizon of this geometry is of $S^5 \times B^3$ topology with the size of S^5 being much bigger than that of B^3 . However, one significant caveat of the construction is that we did not check explicitly that the vacuum geometries (3.13) and (3.26) are solutions of Einstein's equations, but it is rather based on the spirit of [28].

Understanding these bulk descriptions will be beneficial to extract meaningful physics when one also compares with the matrix model side both in analytic, and nowadays, in numerical results.

Acknowledgments

A Thermodynamics of the Vacuum

Here we calculate the entropy and the free energy of the trivial vacuum geometry. We can use the laws of black hole thermodynamics to evaluate the entropy, but for this one needs to know the geometry of the horizon. As it is known and can be easily seen from the metric (3.26) we do not have a simple spherical geometry but we rather have $R \times B^3 \times R \times S^5$. Thus, we have to find the area that this horizon spans such that we can use the Bekenstein-Hawking formula after switching to Einstein frame. Given an arbitrary metric that describes the horizon one can use the following formula

$$\hat{A} = \int_{\mathcal{H}} \sqrt{\det h}, \quad (\text{A.1})$$

where $\mathcal{H}$ is the geometry of the horizon, and h is the induced metric on the horizon. Therefore, we can read this geometry off from (3.26), where it has a manifest $SO(3) \times SO(6)$ symmetry and the horizon has $B^3 \times S^5$ topology¹⁵. Furthermore, we can use the Bekenstein-Hawking formula to evaluate the *physical* entropy as

$$S = e^{-2\phi} \int_{B^3 \times S^5} \sqrt{\det h} = \frac{64\pi^4}{3} e^{-2\phi} \left(\frac{r}{l_s}\right)^5 \left(\frac{z}{l_s}\right)^3 H(r)^2 = \frac{64\pi^4 H(r_0)^{\frac{1}{2}} r_0^5 z_0^3}{3G_{10}}, \quad (\text{A.2})$$

where in the last equality we have used $e^\phi = g_s H(r)^{3/4}$ for the dilaton, and $z \mapsto z_0$ which can be though either as the "extremal" energy scale due to S^2 that corresponds to $\frac{z_0}{\lambda^{1/3}} \sim \frac{\pi}{2}$, or as the distance between the infinite conducting plate and the first conducting plate in figure 3.

Putting everything together we have

$$S \approx N^2 \left(\frac{T}{\lambda^{1/3}}\right)^{3/5}, \quad (\text{A.3})$$

suppressing numbers in front of the right-hand side.

The free energy for the trivial vacuum follows from the first law of thermodynamics and gives

$$\frac{F}{\lambda^{1/3}} \approx \frac{5}{8} N^2 \left(\frac{T}{\lambda^{1/3}}\right)^{8/5}. \quad (\text{A.4})$$

A.1 A note on phase transition

Here we discuss some aspects of Gregory-Laflamme instability [39] regarding this model. The parameter that controls most of the features here is N . A useful interpretation is the number of D0-branes. Then this number controls the regimes of the theory, i.e the regimes where the gravity picture is valid or quantum corrections have to be taken into account. Then, we can imagine a situation where we start investigating a black hole, i.e for simplicity a Schwarzschild-like black hole, when N is large. In this limit, and for specific range of r which is dictated by demanding the curvature not to be large, the

¹⁵We integrate also over z with boundary values $z \in (0, \lambda^{1/3}]$, since at this case we form the maximal area for the horizon.

supergravity description is valid. In addition, decreasing N leads to a critical value $N \sim S$, where the number of D0-branes is the same order as the entropy of the system (S). This rather means that the constituents of the black hole (in this case the D0-branes) give the entropy of the system. Intuitively, this makes sense since each D0-brane carries also a different degree of freedom, resulting in a different (micro)state and thus microstate counting gives the entropy. We note that in the $N \gg S$ regime, we introduce a "needless" amount of degrees of freedom which due to supersymmetry they are in their ground state (energy of fermions cancel the one of bosons). As one approaches the point $S \sim N$ from large N in this specific value both the black hole and the black brane solutions have the same entropy. Further decreasing N leads to a phase transition where the black hole transforms to a gas of D0-branes [40], [41].

The transition from a black hole description to a gas of D0-branes (or vice versa) can be motivated by some thermodynamic arguments as follows. Suppose that we have a system of gas D0-branes and a black hole with the same mass M . Then, we can estimate the entropies of the two systems as [41]

$$S_{\text{BH}} \sim \left(l_{\text{pl}}/L^d \right)^{\frac{1}{D-3}} \sim \frac{M^{\frac{D-2}{D-3}}}{G_N^{D-3}} \quad (\text{A.5})$$

$$S_{\text{gas}} \sim N^{\frac{1}{2}} \left(l_{\text{pl}}^9/L^d \right)^{\frac{1}{D-4}} \left(\frac{E_{LC}}{R} \right)^{\frac{D-2}{2(D-4)}} \sim N^{-\frac{1}{D-4}} \frac{M^{\frac{D-2}{D-4}}}{G_D^{D-4}}. \quad (\text{A.6})$$

Let us fix the dimension of the spacetime to be $D = 10$, which corresponds to $p = 1$ compactified coordinate in $D = 11 - p$ expression. This gives

$$S_{\text{BH}} \sim \frac{M^{\frac{8}{7}}}{G_{10}^7} \quad (\text{A.7})$$

$$S_{\text{gas}} \sim N^{-\frac{1}{6}} \frac{M^{\frac{8}{6}}}{G_{10}^6}. \quad (\text{A.8})$$

From these expressions it is clear that for large N the black hole has greater entropy than the D0 gas for a specific choice of mass. Thus in the large N limit, the black hole is thermodynamically preferred, while for small values of N the gas configuration is preferred.

A different way of thinking for this transition is the following. We remind that N is also related with the boost in the compactified circle, in a sense that the larger the boost the larger the momentum and hence the larger the N . Then starting with small momenta in the eleven-dimensional circle, in the ten-dimensional picture this is a gas of D0-branes. As we boost more, N increases until it reaches a value where the ten dimensional picture is that of a black hole.

However, as we pointed out before the phenomenon we observe is (almost) universal¹⁶ in N , meaning that for every value of N the system behaves the same. The D0-probe approaching from very far away tries to minimize its energy, and it is clear that this occurs on the minima of the potential. Then approaching the horizon, it experiences a repulsive force.

In our case, we can read off the temperature for which we have this instability. This can be understood from the above discussion and equation (A.3). We want to find the "correspondence"

¹⁶Small values of N do not have the same behavior but in this case we should not really trust the gravity description.

point [41], [40], by setting $S \sim N$. This leads to

$$T_c = \frac{\lambda^{1/3}}{N^{5/3}}, \quad (\text{A.9})$$

for the trivial vacuum. This temperature signals when the short of black hole (3.26) has a Gregory-Laflamme instability [39].

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